Exemplar PEER Conclusion

Investigate constant linear motion using primary data (light gate method) –

Data collected

Length of card, l = 10 cm = 0.10 m. Velocity calculated using v = l / t.

Trial	Time, t (s)	Velocity (cm s^{-1})	Velocity (m s^{-1})
1	0.2065	48.43	0.48
2	0.999	10.01	0.10
3	0.2125	47.06	0.47
4	0.1175	85.11	0.85
5	0.1838	54.41	0.54

Velocities in m s^{-1} are given to 2 significant figures because the card length was only measured to 2 significant figures.

Conclusion

P – Point

The velocity of the rider as it passed through the light gate was measured for five different pushes. It ranged from 0.10 m s^{-1} to 0.85 m s^{-1}. The light gate method gave a direct measurement of velocity using v = l / t. However, only one light gate was used, so the velocity was measured at one position on the track only. This investigation could not test whether the velocity stayed constant along the whole track.

E – Evidence

The slowest push (trial 2) took 0.999 s and gave a velocity of 0.10 m s^{-1}. The fastest push (trial 4) took 0.1175 s and gave 0.85 m s^{-1}, which is about 8.5 times greater. Trials 1, 3 and 5 gave similar velocities of 0.48, 0.47 and 0.54 m s^{-1}, but even these differ by up to 16% (0.54 compared with 0.47).

The differences between trials are far larger than the uncertainty in the measurements. The timer was read to 0.1 ms (0.0001 s), which is less than 0.1% of every time recorded. The card length was measured to the nearest millimetre (± 0.1 cm), which is an uncertainty of 1% of 10 cm. The differences of 16% to 750% therefore come from the pushes being different, not from measurement error.

E – Explanation

The timer measures how long the card takes to pass through the beam. Dividing the length of the card by this time gives the average velocity over the length of the card ($v = \Delta s / \Delta t$). The card is short and the time is very small, so this average is very close to the velocity of the rider at the gate. A light gate also removes the reaction-time error that a stopwatch would introduce.

The track was levelled using a spirit level. If the track sloped, the rider would speed up or slow down as it moved, so its velocity would not be the same at every point. The air cushion lifts the rider off the track, which reduces friction, so the rider should not slow down noticeably. With these precautions, the rider should move with constant velocity, which means it covers equal distances in equal times. The velocity measured at the gate should then be the same anywhere along the track. This is a prediction that my data could not test.

The velocity differed between trials because a harder push gives the rider a greater velocity. Each push was different, so each trial gave a different value.

R – Reflection

The main weakness is that one light gate cannot show that the velocity was constant. To improve this, I would use two light gates a known distance apart and measure the velocity at each gate for the same push. I would then calculate the percentage difference between the two velocities. If they agree within the experimental uncertainty (about 1–2%), the result supports the idea that the velocity is constant. It would still not prove it, because the test covers only one track length and a limited range of speeds.

Trial 2 was much slower than the others. I kept it because I had no evidence of a fault, but I would repeat it to check that the time was recorded correctly and the push was not interrupted.

The largest source of uncertainty was the length of the card (1%). I would measure it several times and check that the card is vertical and passes through the beam each time. I would also repeat each push several times at a similar speed, for example by releasing the rider from the same stretched elastic band each time, so that I could calculate a mean and compare the spread of results.

Overall, the method is valid for measuring velocity. The conclusion about constant velocity is limited by the use of a single gate.

Why is this a strong Pconclusion

- **Point:** states the measured values (with units) and what the investigation can and cannot show.

- **Evidence:** quotes specific numbers, a percentage difference and the percentage uncertainty, then uses them to judge whether the differences are real.

- **Explanation:** links the results to $v = \Delta s / \Delta t$, to the levelled air track, and to the idea of constant velocity.

- **Reflection:** names a specific weakness, suggests a specific improvement, and uses the words "supports" and "does not prove".